Product Requirements Document (PRD)

ParkPulse – Smart Parking, Smarter Cities

Version: 1.0
Project Type: IoT + Full Stack Capstone Project

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1. Product Vision

Imagine driving to a busy shopping mall on a weekend. Instead of spending 20–30 minutes searching for a parking space, you simply open the ParkPulse app, reserve an available slot before arriving, navigate directly to it, enter using a QR code, and pay digitally when you leave.

ParkPulse transforms parking from a frustrating experience into a seamless digital journey.

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2. Problem Statement

Urban parking is one of the biggest causes of unnecessary traffic congestion.

Drivers frequently:

- Drive around parking lots looking for empty spaces.
- Waste fuel and valuable time.
- Miss appointments because parking is unavailable.
- Experience stress and frustration.

Parking operators also face challenges:

- No real-time visibility of slot occupancy.
- Manual ticketing and payment.
- Difficulty analyzing parking usage.
- Inefficient utilization of available parking spaces.

3. Product Goal

Build an intelligent parking platform that allows users to:

- Find nearby parking.
- View live parking availability.
- Reserve a parking slot.
- Navigate directly to the slot.
- Enter using QR verification.
- Pay digitally.
- Exit without manual intervention.

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4. Target Users

Drivers

- Daily commuters
- Families visiting malls
- Office employees
- Students
- Airport travelers
- Hospital visitors

Parking Administrators

- Shopping malls
- Colleges
- Hospitals
- Airports
- Office campuses
- Smart city authorities

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5. User Story

Rahul is driving to a shopping mall on a Saturday afternoon.

Normally, he spends nearly 20 minutes looking for parking.

Today, he opens ParkPulse.

The app instantly detects his location and displays nearby parking facilities.

He notices that Phoenix Mall has **18 available parking spaces**.

He taps the parking location.

A live parking map appears showing available and occupied slots.

Rahul reserves Slot A4.

The reservation is held for 15 minutes.

The app generates a QR code.

Google Maps navigates Rahul directly to the parking entrance.

At the entrance, the QR code is scanned.

The gate opens automatically.

Rahul parks in Slot A4.

The sensor detects his vehicle.

The slot immediately changes from **Available** to **Occupied** for everyone using the app.

After shopping, Rahul returns to his vehicle.

The app automatically calculates the parking fee.

Rahul pays using UPI.

The exit gate opens.

When the car leaves, the sensor updates the slot status back to **Available**.

The next driver can now reserve it.

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6. Key Features

Smart Parking Search

Users can search nearby parking lots based on their GPS location.

Features:

- Nearby parking list
- Distance
- Available spaces
- Estimated travel time

Live Parking Availability

Every parking slot updates in real time.

Status:

- Available
- Occupied
- Reserved

Interactive Parking Map

The application displays a visual parking layout.

Example:

Ground Floor

🟩 A1

🟥 A2

🟩 A3

🟨 A4 (Reserved)

🟥 A5

Slot Reservation

Users can reserve parking before arrival.

Reservation duration:

- 15 minutes

If the driver does not arrive within the reservation window, the slot automatically becomes available again.

Google Maps Navigation

Users can navigate directly from their current location to the selected parking area.

QR Code Entry

Every successful reservation generates a unique QR code.

The QR code is scanned at the parking entrance.

Benefits:

- Faster entry
- No paper tickets
- Contactless verification

Digital Payment

Users can pay through:

- UPI
- Credit Card
- Debit Card
- Net Banking

Parking History

Users can view:

- Previous bookings
- Parking duration
- Amount paid
- Parking locations

Notifications

Examples:

- Reservation Confirmed
- Reservation Expiring Soon
- Payment Successful
- Parking Time Reminder

Admin Dashboard

Administrators can monitor:

- Available slots
- Occupied slots
- Reservations
- Revenue
- Parking history
- User activity

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7. Product Workflow

Driver opens ParkPulse

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GPS detects current location

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Nearby parking lots displayed

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Driver selects parking

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Live slot availability displayed

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Driver reserves slot

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QR code generated

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Google Maps navigation

↓

Parking entrance

↓

QR verification

↓

Gate opens

↓

Vehicle parks

↓

Sensor detects vehicle

↓

Firebase updates slot status

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Mobile app updates instantly

↓

Parking timer starts

↓

Digital payment

↓

Vehicle exits

↓

Sensor detects empty slot

↓

Slot becomes available

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8. Functional Requirements

Authentication

- User registration
- Secure login
- Password reset
- Google Sign-In

Parking Search

- Search nearby parking
- Filter by distance
- Filter by availability

Reservation

- Reserve available slot
- Cancel reservation
- Auto-expire reservation

QR Verification

- Generate QR code
- Validate QR code
- Prevent duplicate entry

Parking Monitoring

- Real-time slot updates
- Live parking map

Payment

- Calculate parking fee
- Multiple payment methods
- Payment confirmation

Admin

- Add parking locations
- Manage slots
- View analytics
- Generate reports

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9. Non-Functional Requirements

- Real-time updates within 2 seconds
- Secure user authentication
- Mobile responsive interface
- 99% system availability
- Scalable cloud architecture
- Easy-to-use interface

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10. Hardware Requirements

- ESP32
- Ultrasonic Sensor
- IR Sensor (Optional)
- QR Scanner
- Wi-Fi Network
- LED Indicators

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11. Software Requirements

Frontend

- React.js / Flutter

Backend

- Node.js
- Express.js

Database

- Firebase Realtime Database

Authentication

- Firebase Authentication

Maps

- Google Maps API

Payment

- UPI Payment Gateway

QR

- QR Code Generator

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12. System Architecture

Vehicle

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Ultrasonic Sensor

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ESP32

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Wi-Fi

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Firebase Cloud

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Backend API

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React / Flutter App

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Driver & Admin Dashboard

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13. Success Metrics

- Parking search time reduced by 80%
- Average booking completed in under 30 seconds
- Real-time updates delivered in under 2 seconds
- Reduction in parking congestion
- Increased parking space utilization
- Improved user satisfaction

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14. Future Scope

- Parking demand prediction using machine learning
- Automatic Number Plate Recognition (ANPR)
- Electric Vehicle charging slot booking
- Dynamic pricing based on demand
- Voice assistant support
- Smart city integration
- Multi-city parking network
- Offline sensor synchronization
- Predictive parking recommendations

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15. Why ParkPulse?

ParkPulse is more than a parking application—it is an intelligent mobility platform that connects IoT devices, cloud computing, and modern web technologies to simplify urban parking. By enabling real-time visibility, reservations, contactless entry, and digital payments, ParkPulse reduces congestion, saves time, and enhances the overall parking experience. Its scalable architecture also makes it suitable for future integration into smart city ecosystems.